

Recuperare informazioni per le meta-analisi

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recuperare informazioni descrittive



① media: 49 anni

② s.d.: 27.2 anni

mediana ?
quartili ?

oppure: mediana 44, primo quartile 41, terzo quartile 74.

Media? Deviazione standard?

recuperare informazioni descrittive / 1 of 2

Possiamo dedurre la media μ dalla mediana, dai quartili e dal range?

$$\mu \approx \frac{a + 2m + b}{4} + \frac{a - 2m + b}{4n} \approx \frac{a + 2m + b}{4}$$

$$\mu \approx \frac{a + 2Q_1 + 2m + 2Q_3 + b}{8}$$

$$\mu \approx \frac{Q_1 + m + Q_3}{3}$$

recuperare informazioni descrittive / 2 of 2

Possiamo dedurre la deviazione standard σ dalla mediana, dai quartili e dal range?

$$\sigma \approx \frac{b - a}{\xi(n)}$$

$$\sigma \approx \frac{Q_3 - Q_1}{\eta(n)}$$

$$\sigma \approx \frac{1}{2} \left(\frac{b - a}{\xi(n)} + \frac{Q_3 - Q_1}{\eta(n)} \right)$$

Table 1: Numerical values of $\xi(n)$ when $n \leq 50$, according to Wan et al.

n	$\xi(n)$	n	$\xi(n)$	n	$\xi(n)$	n	$\xi(n)$	n	$\xi(n)$
1	0	11	3,173	21	3,778	31	4,113	41	4,341
2	1,128	12	3,259	22	3,819	32	4,139	42	4,361
3	1,693	13	3,336	23	3,858	33	4,165	43	4,379
4	2,059	14	3,407	24	3,895	34	4,189	44	4,398
5	2,326	15	3,472	25	3,931	35	4,213	45	4,415
6	2,534	16	3,532	26	3,964	36	4,236	46	4,433
7	2,704	17	3,588	27	3,997	37	4,259	47	4,450
8	2,847	18	3,640	28	4,027	38	4,280	48	4,466
9	2,970	19	3,689	29	4,057	39	4,301	49	4,482
10	3,078	20	3,735	30	4,086	40	4,322	50	4,498

Table 2: Numerical values of $\eta(n)$ when $n \leq 50$, according to Wan et al.

n	$\eta(n)$	n	$\eta(n)$	n	$\eta(n)$	n	$\eta(n)$	n	$\eta(n)$
1	0,990	11	1,307	21	1,327	31	1,334	41	1,338
2	1,144	12	1,311	22	1,328	32	1,334	42	1,338
3	1,206	13	1,313	23	1,329	33	1,335	43	1,338
4	1,239	14	1,316	24	1,330	34	1,335	44	1,338
5	1,260	15	1,318	25	1,330	35	1,336	45	1,339
6	1,274	16	1,320	26	1,331	36	1,336	46	1,339
7	1,284	17	1,322	27	1,332	37	1,336	47	1,339
8	1,292	18	1,323	28	1,332	38	1,337	48	1,339
9	1,298	19	1,324	29	1,333	39	1,337	49	1,339
10	1,303	20	1,326	30	1,333	40	1,337	50	1,340

Un esempio



- the Reference Data Set
 - dal 2003 al 2014, King's College London Dental Institute
- LL8Gf
 - Lower Left Third Molar
 - Stage G female
 - 18 years old
 - caucasian UK

Un esempio

<https://www.dentalage.co.uk/rds-uk-caucasian/>

$$n = 114$$

$$\mu \approx \frac{a + 2Q_1 + 2m + 2Q_3 + b}{8}$$

$$\frac{14.9 + 34.2 + 36.2 + 39.9 + 22.8}{8}$$

$$\mu \approx 18.5$$

0th%ile	0.5%ile	5%ile	10%ile
14.96	15.73	15.97	16.32
25%ile	50%ile	75%ile	90%ile
17.07	18.13	18.99	20.64
95%ile	99.5%ile	100th%ile	
21.35	21.91	22.78	

Un esempio

<https://www.dentalage.co.uk/rds-uk-caucasian/>

$$n = 114$$

$$\sigma \approx \frac{1}{2} \left(\frac{b - a}{\xi(n)} + \frac{Q_3 - Q_1}{\eta(n)} \right)$$

$$\frac{1}{2} \left(\frac{22.8 - 14.9}{4.55} + \frac{19.0 - 17.1}{1.35} \right)$$

$$\sigma \approx 1.6$$

0th%ile	0.5%ile	5%ile	10%ile
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14.96	15.73	15.97	16.32
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25%ile	50%ile	75%ile	90%ile
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17.07	18.13	18.99	20.64
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95%ile	99.5%ile	100th%ile
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21.35	21.91	22.78
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Un esempio

$$\mu \approx 18.5$$

$$\sigma \approx 1.6$$

TDS	n	mean	sd
LL8Gf	114	18.25	1.63

$$\sigma = \frac{b-a}{\xi(n)}$$

$$\xi(n) \approx \begin{cases} 2 \cdot \Phi^{-1} \left(\frac{n-0.375}{n+0.25} \right) + 0.0197 \log(n) - 0.0626 & \text{if } 2 \leq n \leq 50 \\ 2 \cdot \Phi^{-1} \left(\frac{n-0.375}{n+0.25} \right) & \text{if } n > 50 \end{cases}$$

$$\sigma = \frac{Q_3 - Q_1}{\eta(n)}$$

$$\eta(n) \approx \begin{cases} 2 \cdot \Phi^{-1} \left(\frac{0.75n-0.125}{n+0.25} \right) + \exp \left(\frac{n}{-2.882-0.231 \cdot n} \right) & \text{if } 2 \leq n \leq 50 \\ 2 \cdot \Phi^{-1} \left(\frac{0.75n-0.125}{n+0.25} \right) & \text{if } n > 50 \end{cases}$$

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2*qnorm((n-0.375)/(n+0.25))
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2*qnorm((0.75*n-0.125)/(n+0.25))
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Commentary to Wan et al. (2014):
Estimating the standard deviation
from the sample size and range or quartiles

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Abstract

This short note proposes two additive corrections to a pair of relations published in Wan et al. [WWLT14] in order to extend them to a 'small sample size' condition. In particular we focus the interest on the possibility to provide an estimate to the sample standard deviation σ when knowing only the sample size n , the range $[a, b]$ and/or the quartiles Q_1, Q_3 of some data. Our results allow to explicitly compute σ , for instance with software R or any spreadsheet, for any sample size $n \geq 2$. R codes and data are publicly available on <https://github.com/MassimoBorelli/sd>

